

2014-2015 – Licence 3 – Macro

Lecture 9 : The Liquidity Trap

Franck Portier

franck.portier@TSE-fr.eu

Toulouse School of Economics

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Disclaimer

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0. Introduction

- ▶ What is the Liquidity Trap : a situation in which :
 - × the nominal interest rate i is zero,
 - × the economy is in a recession,
 - × monetary policy is inefficient because it cannot reduce i any further
- ▶ Why can't i be negative : it is possible on bonds (i is a contractual arrangement)
- ▶ But then people will hold money, not bonds, because the nominal interest rate on money is zero
- ▶ It is not *technically* possible to make it negative (unless one uses *melting money*)
- ▶ Therefore demand for bond would be zero if i was negative on bonds

0. Introduction

- ▶ We first briefly review the liquidity trap in IS-LM
- ▶ Then we look at facts to show that it is not a theoretical curiosity :
 - × the US in the 1930's (Great Depression),
 - × Japan in the 1990's (Lost Decade),
 - × the US and France after 2008 (Great Recession).
- ▶ Then we construct a micro-founded model that gives a rigorous treatment of the liquidity trap.
- ▶ We will see in particular that expectations are crucial to understand what is going on.

1. IS-LM

Assumptions

- ▶ Prices are fixed
- ▶ Output is determined by demand
- ▶ Economic agents make portfolio decisions and decide to hold money and bonds.

1. IS-LM

IS

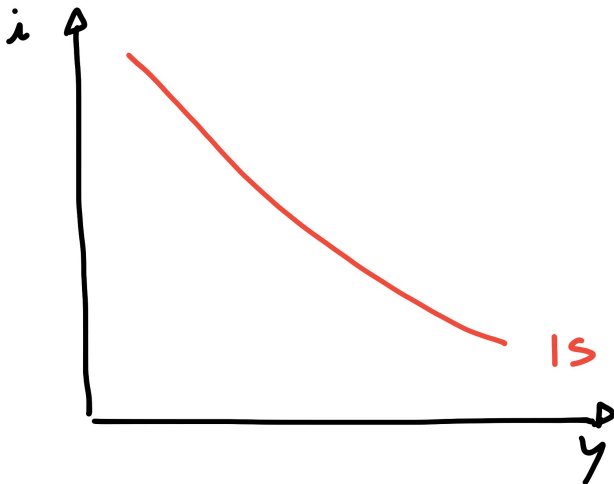
- ▶ $C = C(Y - T)$, $C' > 0$
- ▶ $I = I(i)$, $I' < 0$
- ▶ $i = r + \pi$, where π is inflation
- ▶ G is given
- ▶ $G + B^G = T + M/P$, T , M and B^G given, but satisfies the Gvt budget constraint
- ▶ Equilibrium : Income Y creates demand that is equal to income :

$$C + I + G = Y \quad (IS)$$

- ▶ IS is downward sloping in the plane (Y, i)

1. IS-LM

FIGURE 1 : The IS curve



1. IS-LM

LM

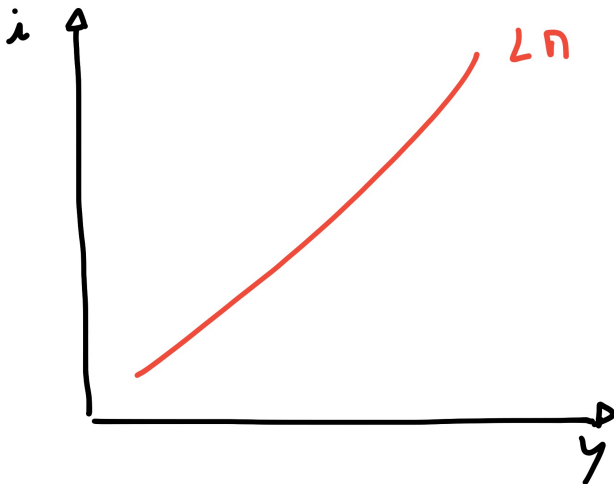
- ▶ Money supply is given M/P
- ▶ Real money demand is $L = L(Y, i)$ with $L_Y > 0$ and $L_i < 0$
- ▶ $L(Y, i)$ is derived from an arbitrage between money and bonds :
 - × when income is high, money is needed for transaction, so that economic agents use more money (and less bonds) as a mean of savings,
 - × when i is high, the opportunity cost of holding money is high, so that agents hold less money (and more bonds)
 - × Equilibrium (on the money market) :

$$L(Y, i) = M/P \quad (LM)$$

- × LM is upward sloping in the plane (Y, i)

1. IS-LM

FIGURE 2 : The LM curve



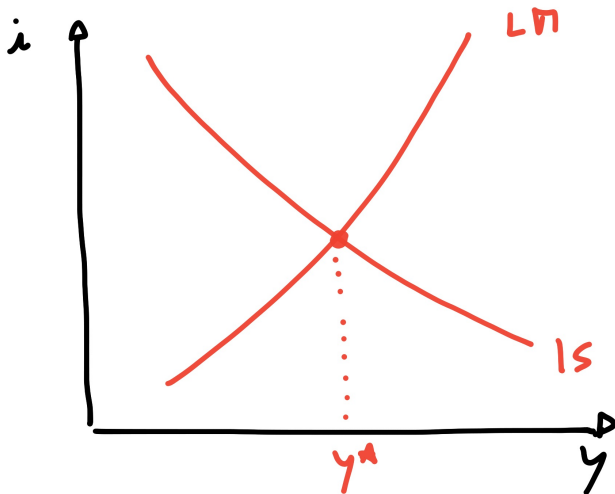
IS-LM

Equilibrium

- ▶ Equilibrium is (Y^*, i^*) such that
 - × Income generate a demand of good equal to income
 - × Money market clears
 - × By Walras law, the bond market also clears
- ▶ The equilibrium is at the intersection of IS and LM

1. IS-LM

FIGURE 3 : Equilibrium in IS-LM



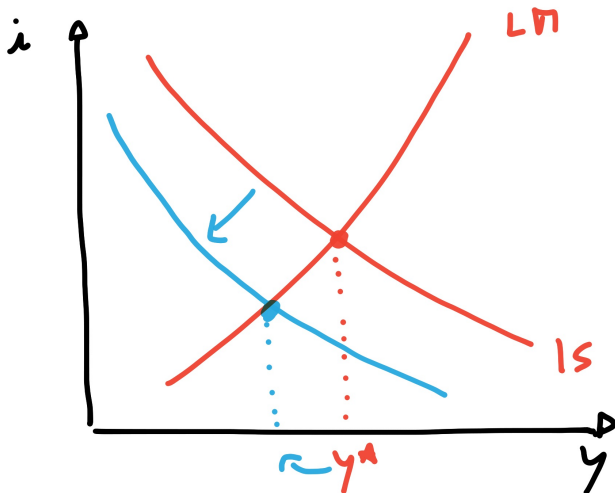
IS-LM

Negative demand shock and the role of expansionary policy

- ▶ Suppose that aggregate demand goes down (for example autonomous investment goes down following a stock market crash)
- ▶ This is likely to create a recession
- ▶ Expansionary monetary policy can restore a high Y equilibrium :
 - × Expansionary monetary policy reduced i
 - × a drop in i increases I
 - × the increase in I has a multiplier effect

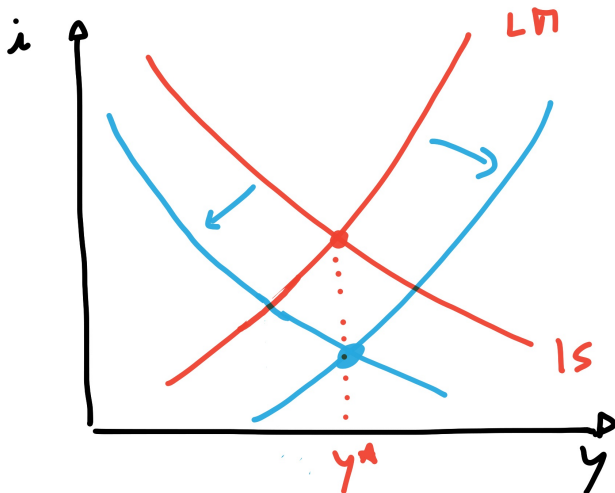
1. IS-LM

FIGURE 4 : A negative shock to aggregate demand



1. IS-LM

FIGURE 5 : Expansionary monetary policy



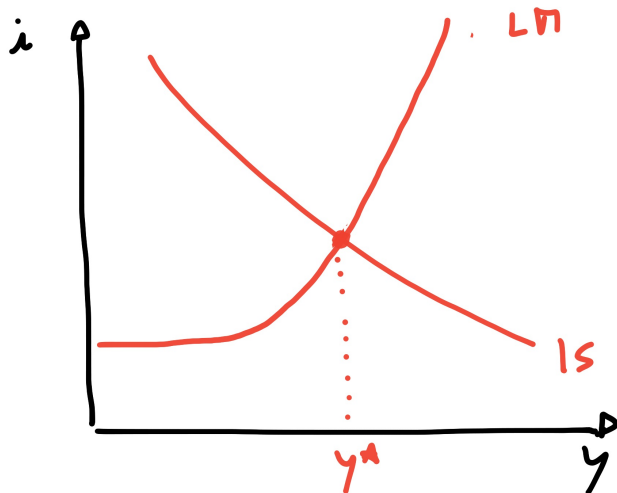
IS-LM

The liquidity trap

- ▶ Assume there is a level of i that is so low (say $i \approx 0$) such that
 - × the demand for bonds is very low (because the return is almost zero)
 - × an increase in money supply cannot decrease any more i because all the money distributed is kept in cash and -is not used to buy bonds
- ▶ In that case, the elasticity of L with respect to i is zero when i is at the *Zero Lower Bound*.
- ▶ Monetary policy cannot reduce i , and is therefore inefficient.
- ▶ If the government cannot run a deficit (because public debt is already high), then there is no possibility to eliminate the recessionary effect of a negative demand shock.

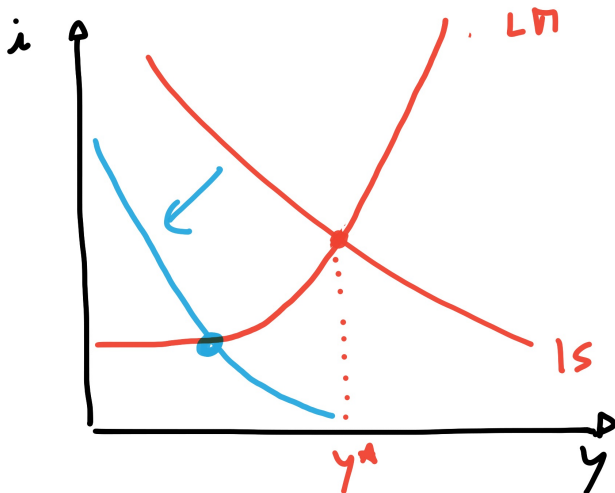
1. IS-LM

FIGURE 6 : The liquidity trap



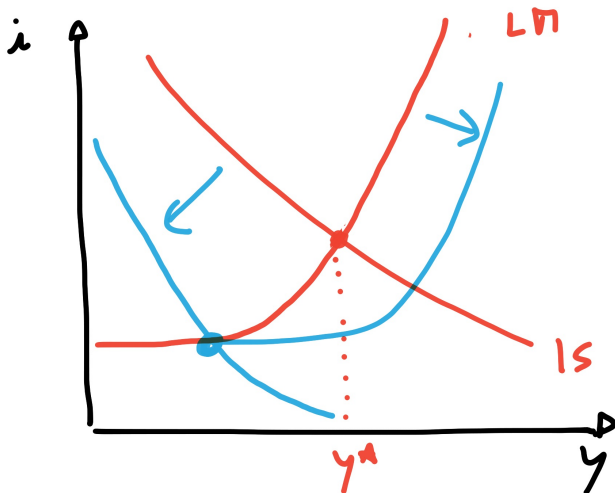
1. IS-LM

FIGURE 7 : The liquidity trap



1. IS-LM

FIGURE 8 : The liquidity trap



2. Facts

- ▶ We look at facts to show that the liquidity trap is not a theoretical curiosity :
 - × the US in the 1930's (Great Depression),
 - × Japan in the 1990's (Lost Decade),
 - × the US and France after 2008 (Great Recession).

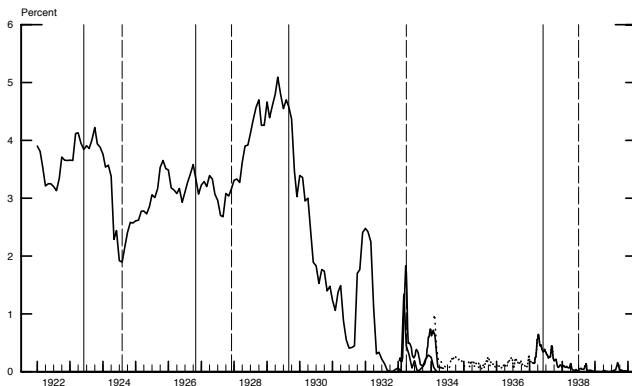
2. Facts

- ▶ In all cases, we observe a similar pattern :
 - × Asset prices (stock market) crashes
 - × nominal interest rate is almost zero
 - × there is a recession in output
 - × unemployment is high
 - × there is a initial period of deflation (not prolonged for the 2008 recession)

2. Facts

The US Great Depression

FIGURE 9 : Short-term interest rates

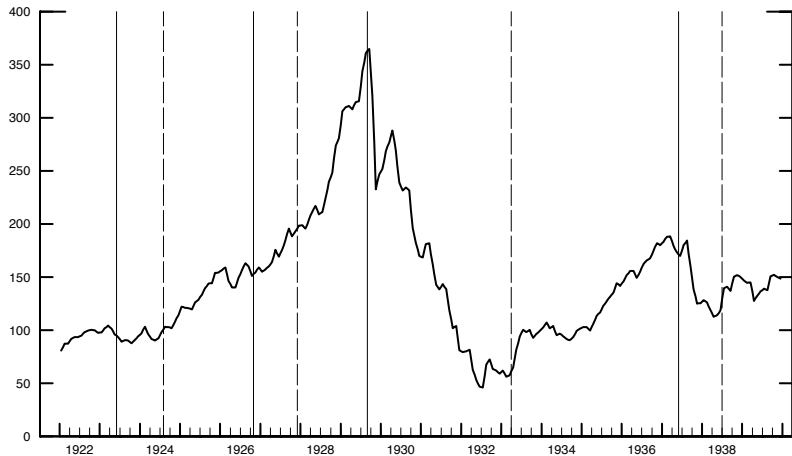


The solid line denotes the yield on new issues of 3-month Treasury bills or equivalents. The dotted lines (from 1934 to 1937) reflect yields on new issues of 6- and 9-month Treasury bills.

2. Facts

The US Great Depression

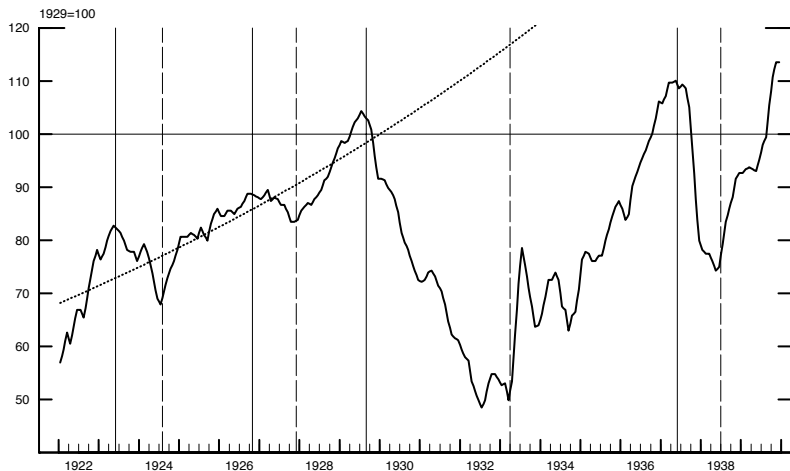
FIGURE 10 : Equity prices (Dow Jones Index)



2. Facts

The US Great Depression

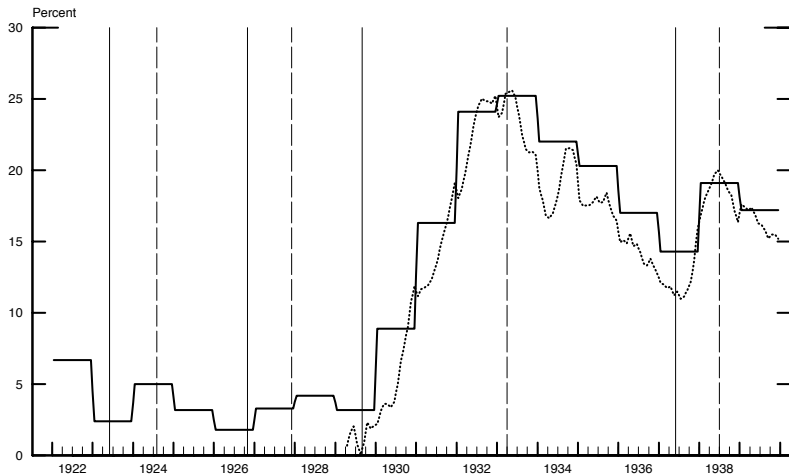
FIGURE 11 : Industrial production



2. Facts

The US Great Depression

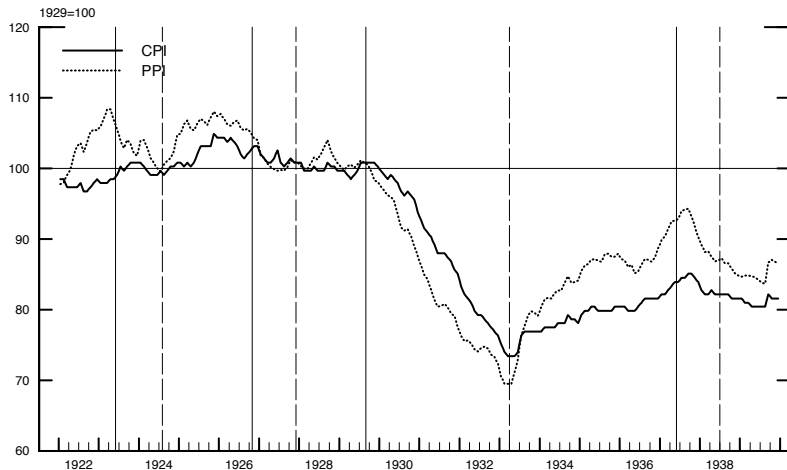
FIGURE 12 : Unemployment rate



2. Facts

The US Great Depression

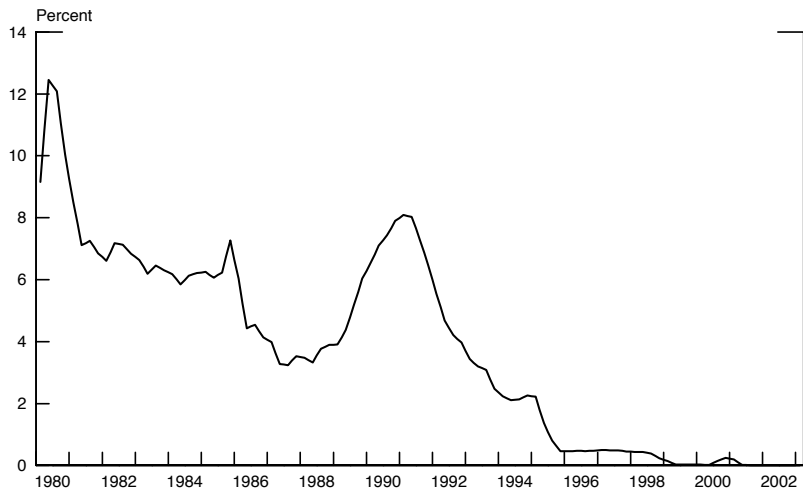
FIGURE 13 : Consumer and Producer Prices



2. Facts

The Japan's lost decade of the 1990s

FIGURE 14 : Overnight Interest rate



2. Facts

The Japan's lost decade of the 1990s

FIGURE 15 : Equity prices (Nikkei Index)



2. Facts

The Japan's lost decade of the 1990s

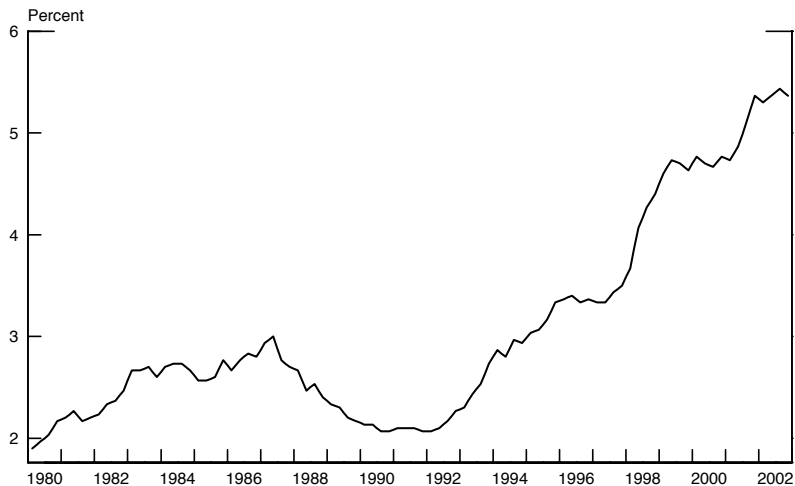
FIGURE 16 : Industrial production



2. Facts

The Japan's lost decade of the 1990s

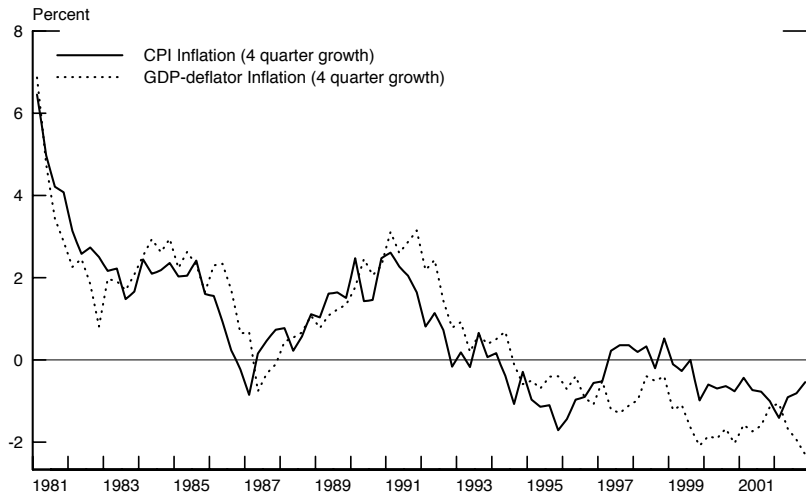
FIGURE 17 : Unemployment rate



2. Facts

The Japan's lost decade of the 1990s

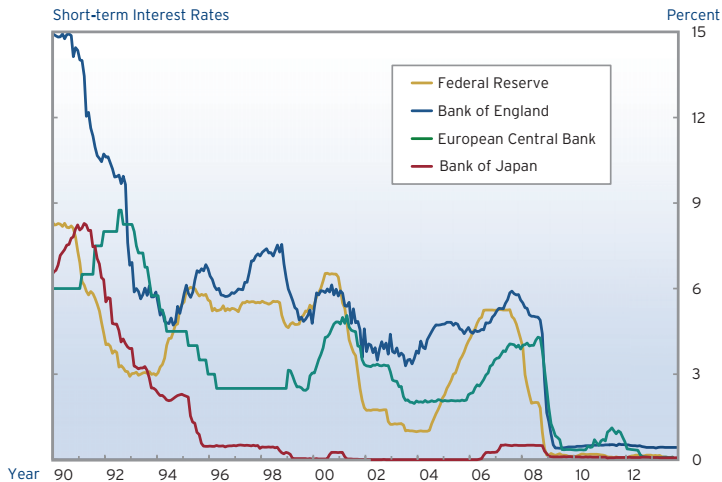
FIGURE 18 : Inflation (CPI and GDP deflator)



2. Facts

The Great Recession

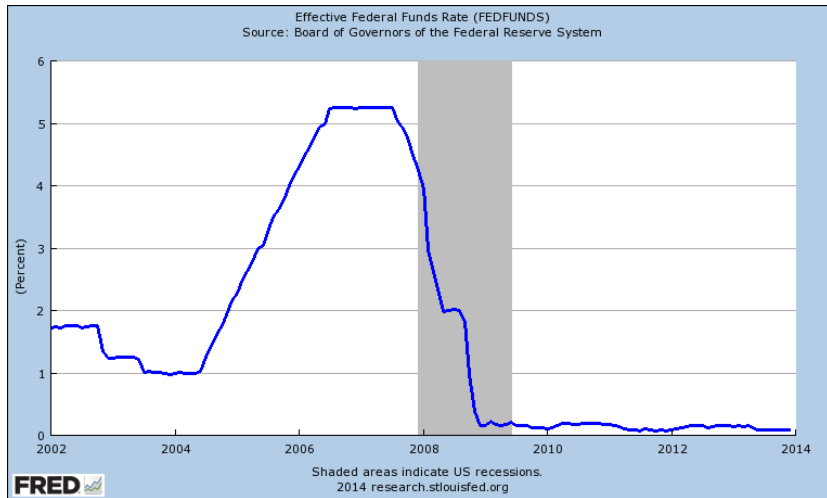
FIGURE 19 : Short term interest rates



2. Facts

The US Great Recession

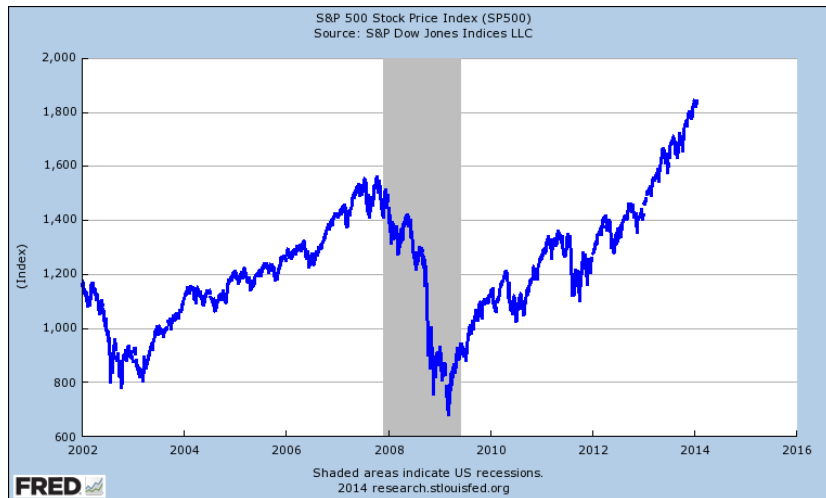
FIGURE 20 : Federal Fund Rate



2. Facts

The US Great Recession

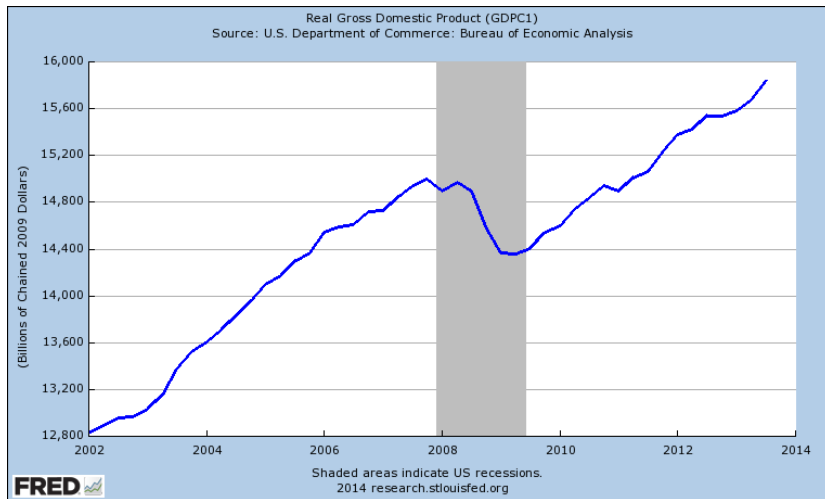
FIGURE 21 : Equity prices (S&P500 Index)



2. Facts

The US Great Recession

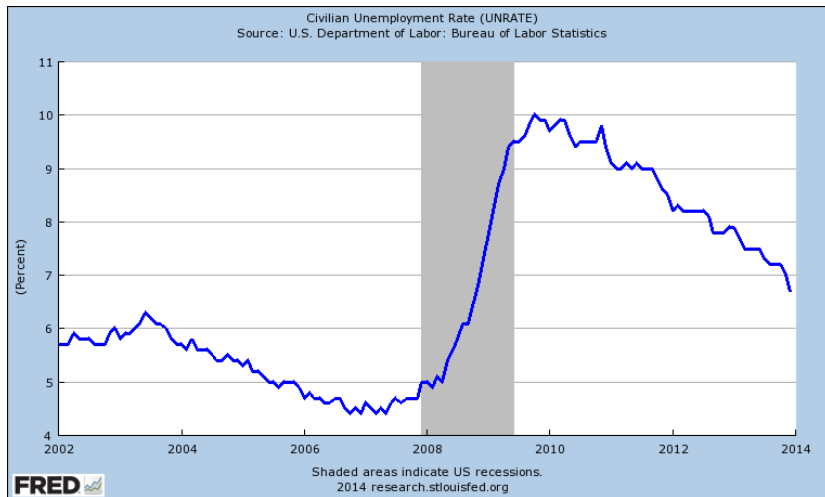
FIGURE 22 : Real GDP



2. Facts

The US Great Recession

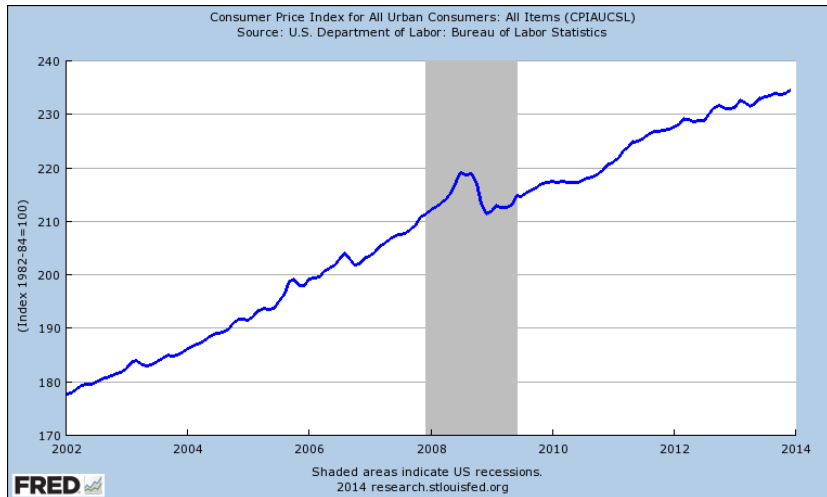
FIGURE 23 : Unemployment rate



2. Facts

The US Great Recession

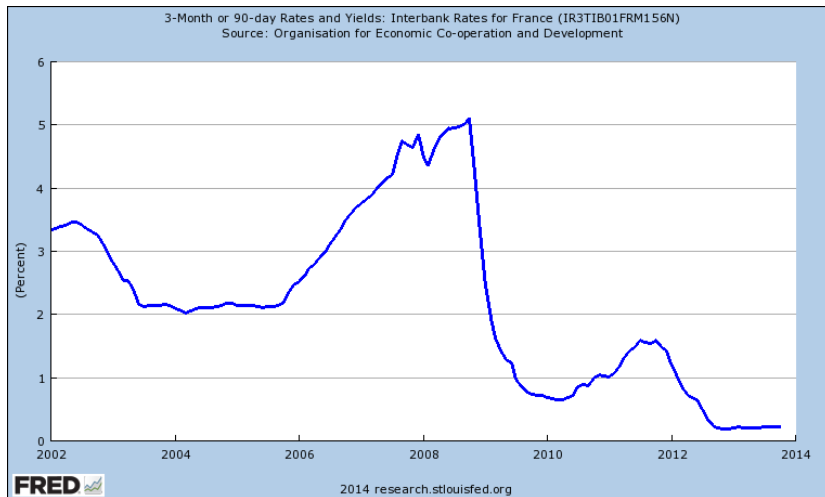
FIGURE 24 : Consumer Price Index)



2. Facts

The French Great Recession

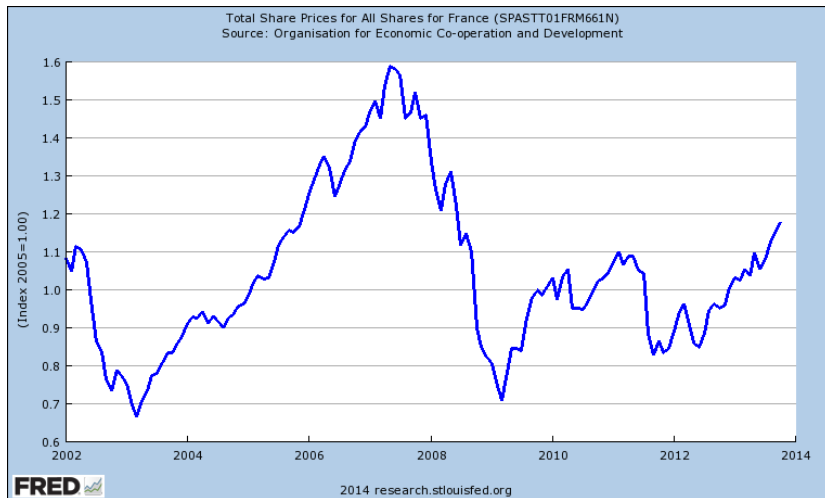
FIGURE 25 : 90-day Interbank Rate



2. Facts

The French Great Recession

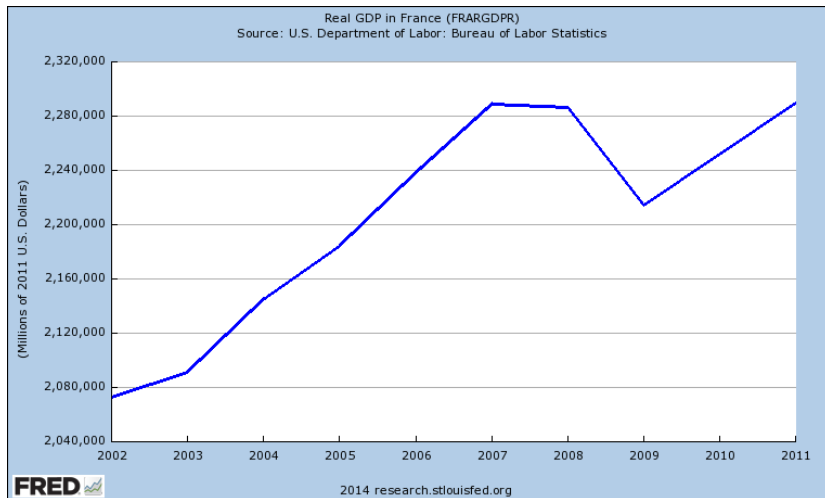
FIGURE 26 : Equity prices (Total Share Price Index)



2. Facts

The French Great Recession

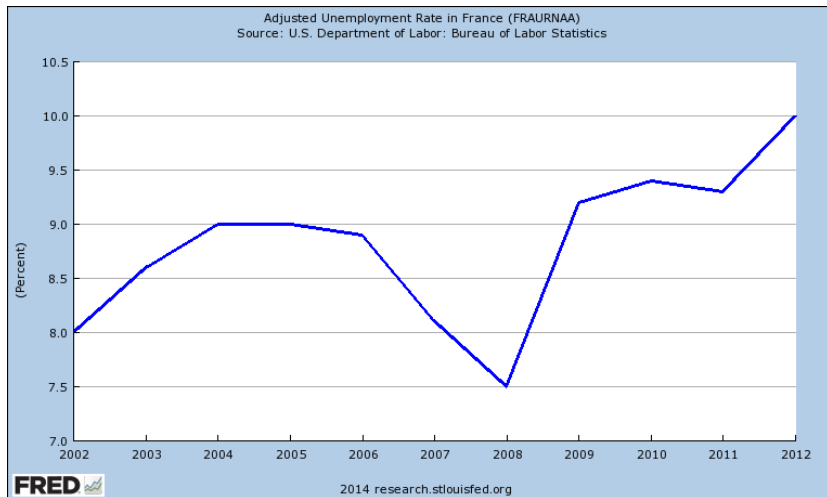
FIGURE 27 : Real GDP



2. Facts

The French Great Recession

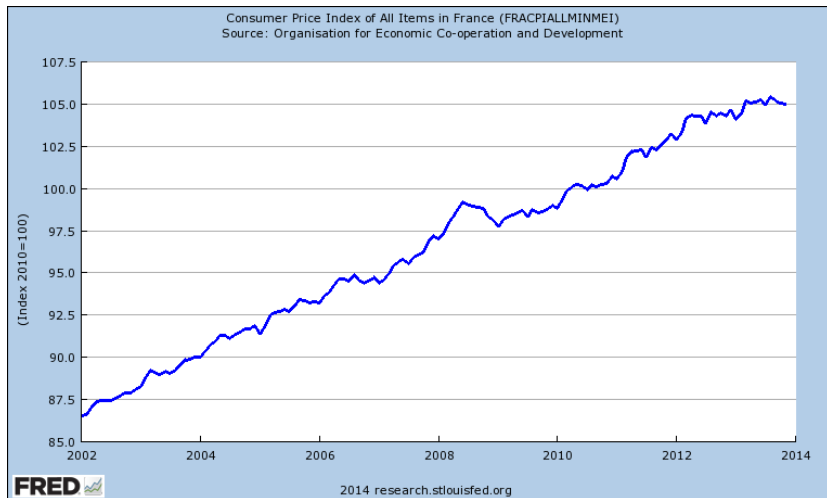
FIGURE 28 : Unemployment rate



2. Facts

The French Great Recession

FIGURE 29 : Consumer Price Index



3. A Microfounded Model

The model

- ▶ A model proposed by PAUL KRUGMAN (1998)
- ▶ It gives a general equilibrium with micro-foundations and dynamics to IS-LM
- ▶ Deterministic economy
- ▶ One representative household

$$U = \sum_{t=0}^{\infty} \beta^t \log c_t$$

with $1 > \beta > 0$.

- ▶ Endowment economy : y_t given
- ▶ The good is perishable
- ▶ Two assets : Money M_t and Bonds B_t

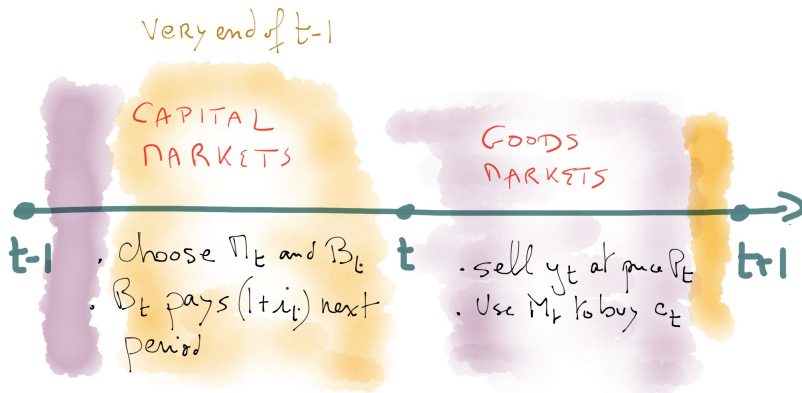
3. A Microfounded Model

The model

- Cash in Advance constraint

$$P_t c_t \leq M_t$$

- Timing :



3. A Microfounded Model

Solving for the household optimal behavior

- ▶ The Hh maximizes utility under a sequence of period t budget constraint and cash in advance constraint :

$$\begin{aligned} B_{t+1} + M_{t+1} + P_t c_t &\leq (1 + i_t) B_t + M_t + P_t y_t + X_t & (\lambda_t) \\ P_t c_t &\leq M_t & (\mu_t) \end{aligned}$$

- ▶ X is newly created money distributed to Hh
- ▶ λ and μ are positive or nul Lagrange multipliers

3. A Microfounded Model

Solving for the household optimal behavior

- The Lagrangian is

$$\begin{aligned}\mathcal{L} = & \sum_{t=0}^{\infty} \beta^t \log c_t \\ & + \sum_{t=0}^{\infty} \lambda_t ((1+i_t)B_t + M_t + P_t y_t + X_t - B_{t+1} - M_{t+1} - P_t C_t) \\ & + \sum_{t=0}^{\infty} \mu_t (M_t - P_t C_t)\end{aligned}$$

3. A Microfounded Model

Solving for the household optimal behavior

- and FOC are $\forall t \geq 0$

$$\begin{array}{lll} \text{w.r.t. } c_t & : & \beta^t u'(c_t) = (\mu_t + \lambda_t) P_t \\ \text{w.r.t. } B_{t+1} & : & \lambda_t = (1 + i_{t+1}) \lambda_{t+1} \\ \text{w.r.t. } M_{t+1} & : & \lambda_t = \lambda_{t+1} + \mu_{t+1} \\ \text{slackness} & : & \mu_t (M_t - P_t c_t) = 0 \\ & & \mu_t \geq 0 \end{array}$$

- There is also a slackness condition for the budget constraint. I do not write it as it will always bind.
- Proof : Assume $\lambda_t = 0$. Then the third FOC implies $\lambda_{t+1} + \mu_{t+1} = 0$. As multipliers are positive or nul, this implies $\lambda_{t+1} = \mu_{t+1} = 0$. Therefore, the first FOC implies $\beta^t u'(c_t) = 0$. This is not possible because $u'(c_t) = 1/c_t > 0$. Therefore $\lambda_t = 0$ is not possible, so that $\lambda_t > 0$ for all t .

3. A Microfounded Model

Solving for the household optimal behavior

- Which gives (*see details in class*)

$$1 + i_t = \underbrace{\left(\frac{u'(c_t)}{\beta u'(c_{t+1})} \right)}_{1+r_t} \left(\frac{P_{t+1}}{P_t} \right)$$

$$P_t c_t = M_t \quad \text{if } i_t > 0$$

- If $i_t = 0$, then $\mu_t = 0$: the CIA constraint may not be binding : Hh is indifferent between money and bonds.

3. A Microfounded Model

Equilibrium with flexible prices

- ▶ All agents are identical (representative household) $\rightsquigarrow$ The equilibrium on the bond market is

$$B_t = 0$$

- ▶ Money market equilibrium :

$$M_{t+1} = M_t + X_t$$

- ▶ Good market equilibrium

$$c_t = y_t$$

- ▶ Remark : the equilibrium quantities can be determined without using the FOC
- ▶ The FOC are determining *equilibrium prices*, i.e. the prices P and i such that markets clear.

3. A Microfounded Model

Equilibrium with flexible prices

- ▶ In what follows, we restrict to a specific case :
 - × for $t \geq 1$, $y_t = y^*$, $M_t = M^*$.
 - × Period 0 : y , M .
- ▶ We can solve the model in the following way
 - × Solve for period 1 and onwards
 - × Solve for period 0

3. A Microfounded Model

Equilibrium with flexible prices - Period 1 and onwards

- ▶ $c_t = y_t = y^*$ implies

$$1 + r^* = \frac{1}{\beta} > 1$$

such that $r^* > 0$

- ▶ Let us guess an equilibrium in which the CIA constraint binds :

$$P^* = \frac{M^*}{y^*}$$

- ▶ In such a case,

$$1 + i_t = \frac{(1 + r_t)P_{t+1}}{P_t}$$

- ▶ implies

$$1 + i^* = \frac{(1 + r^*)P^*}{P^*} = 1 + r^*$$

- ▶ Therefore $i^* = r^* > 0$
- ▶ Because $i^* > 0$, the CIA binds, so that our guess was correct.

3. A Microfounded Model

Equilibrium with flexible prices - Period 0

- ▶ The equilibrium in period $t \geq 1$ is therefore y^* ,
 $i^* = r^* = 1/\beta$, $P^* = M^*/y^*$
- ▶ The equilibrium (i, P) in period 0 is given by the equations :

$$i = \frac{y^*}{\beta y} \frac{P^*}{P} - 1$$

$$P = \frac{M}{y} \quad \text{if } i > 0$$

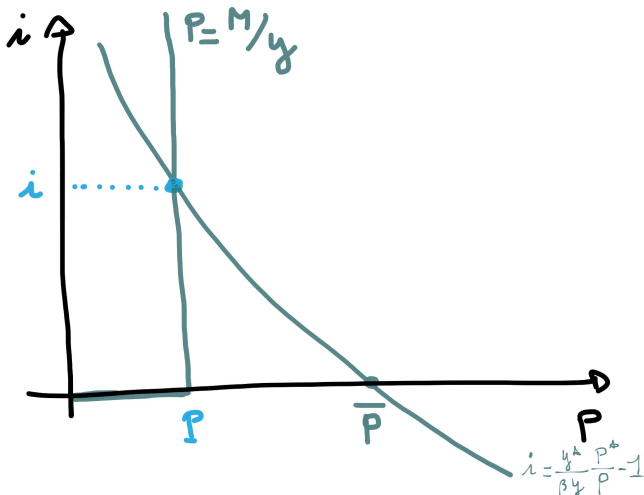
$$P = \bar{P} \quad \text{if } i = 0$$

- ▶ where we have to define $\bar{P}$
- ▶ $\frac{P^*}{P} - 1$ represents expected inflation

3. A Microfounded Model

Equilibrium with flexible prices - Period 0

FIGURE 30 : Equilibrium in period 0 - Flexible prices



3. A Microfounded Model

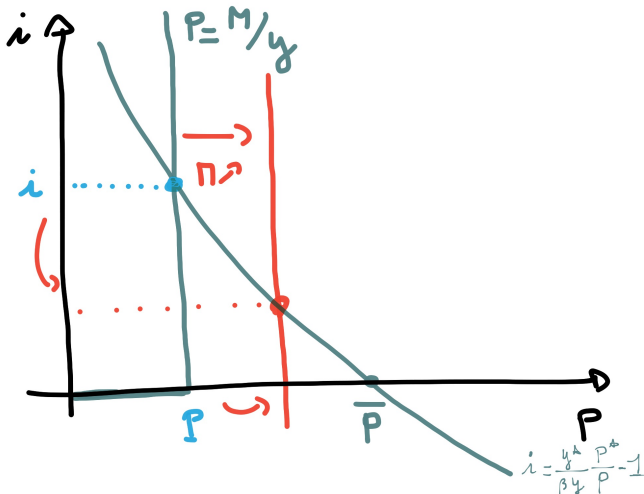
Equilibrium with flexible prices - Period 0

- ▶ How does the model works ?
- ▶ Suppose an increase in money supply today M
- ▶ Prices today P increase
- ▶ As P^* is given, inflation is reduced.
- ▶ Because consumption is independent of monetary policy, $c = y$ and the real interest rate stays constant
- ▶ Therefore, the decrease in inflation decreases i
- ▶ Note that money is neutral
- ▶ Important : the above analysis is true as long as $P > \bar{P}$, such that $i > 0$ and the CIA binds.

3. A Microfounded Model

Equilibrium with flexible prices - Period 0

FIGURE 31 : Equilibrium in period 0 - Flexible prices

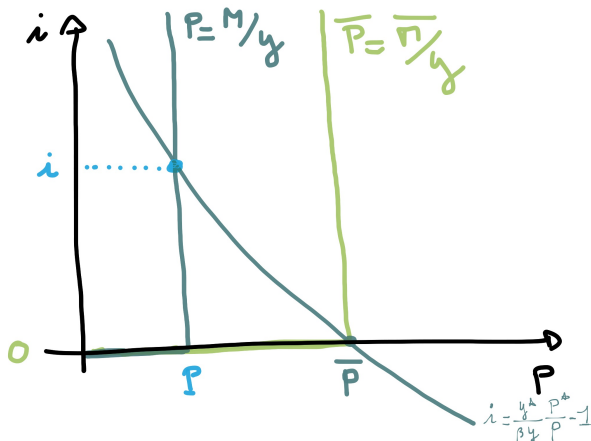


3. A Microfounded Model

Equilibrium with flexible prices - Period 0 - The liquidity trap

- Consider the money supply $\bar{M}$ such that $P = \bar{P}$ and $i = 0$

FIGURE 32 : Equilibrium in period 0 - Flexible prices - $M = \bar{M}$



3. A Microfounded Model

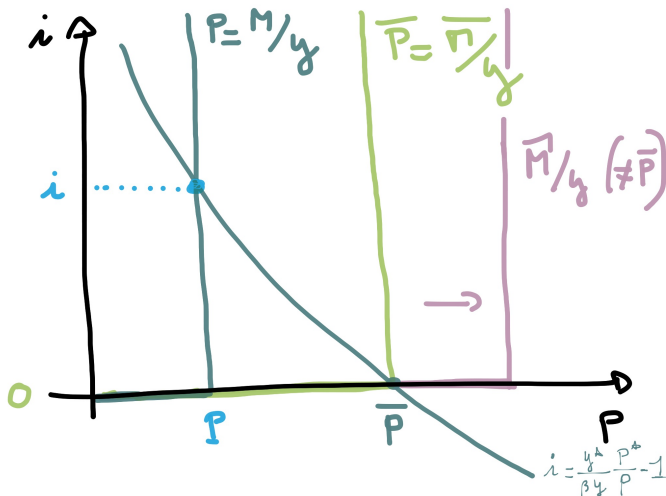
Equilibrium with flexible prices - Period 0 - The liquidity trap

- ▶ Consider the money supply $\overline{M}$ such that $P = \overline{P}$ and $i = 0$
- ▶ What does happen if M increases above $\overline{M}$ to the level $\hat{M}$
- ▶ If P increases above $\overline{P}$, then $i < 0$ which is not possible.
- ▶ Therefore, we must have $i = 0$, and then $P = \overline{P}$
- ▶ As $i = 0$, the CIA does not bind
- ▶ All the increase in money supply is kept in money (not put in bonds), without any impact on the nominal interest rate i .

3. A Microfounded Model

Equilibrium with flexible prices - Period 0 - The liquidity trap

FIGURE 33 : Equilibrium in period 0 - Flexible prices - $M = \hat{M}$



3. A Microfounded Model

Equilibrium with flexible prices - Period 0 - The liquidity trap

- ▶ Remark : In the liquidity trap with flexible prices, money cannot reduce inflation below $\frac{P^*}{\bar{P}}$ (which is actually *deflation*)
- ▶ But money is always neutral ... so that we do not care much about the liquidity trap.
- ▶ Things will be different if we assume, as in IS-LM, that prices are fixed in period 0.

3. A Microfounded Model

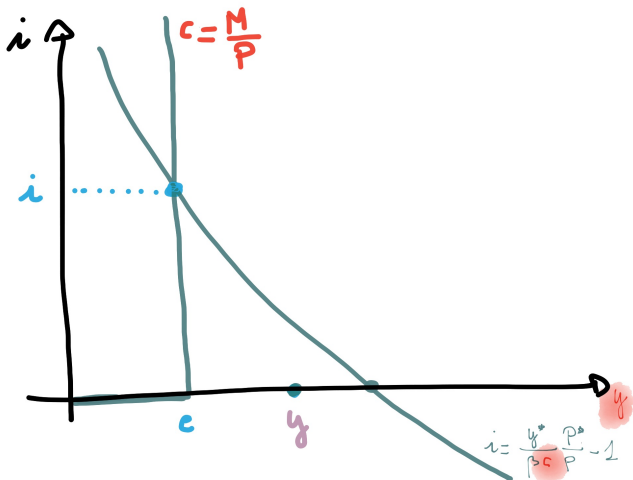
Equilibrium with fix prices - Period 0

- ▶ Assume first that $i > 0$, so that the economy is not in the liquidity trap.
- ▶ As the CIA constraints binds, we have $Pc = M$, with $c \leq y$
- ▶ Monetary policy now determines c as P is fixed

3. A Microfounded Model

Equilibrium with flexible prices - Period 0

FIGURE 34 : Equilibrium in period 0 - Fix prices



3. A Microfounded Model

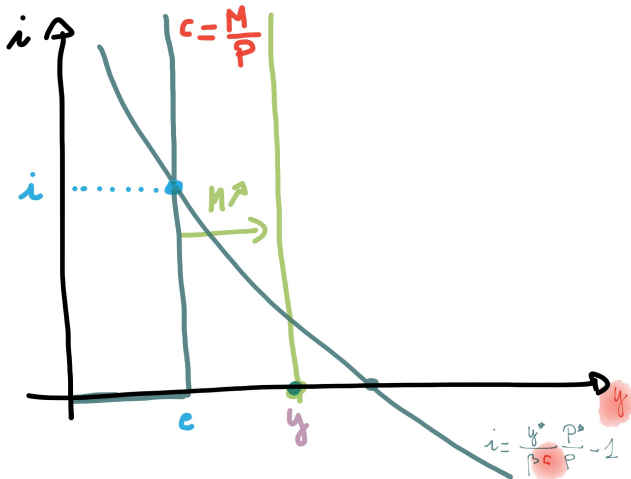
Equilibrium with fix prices - Period 0

- ▶ The economy is in a configuration similar to IS-LM
- ▶ As $c < y$, the equilibrium is inefficient (an amount $y - c$ of goods is wasted)
- ▶ Assume an increase in M , for given M^* : to clear the money market, i needs to go down. As inflation is fixed, there real interest rate must decrease, which is achieved by an increase in c .

3. A Microfounded Model

Equilibrium with fix prices - Period 0

FIGURE 35 : Equilibrium in period 0 - Fix prices - Monetary policy is efficient



3. A Microfounded Model

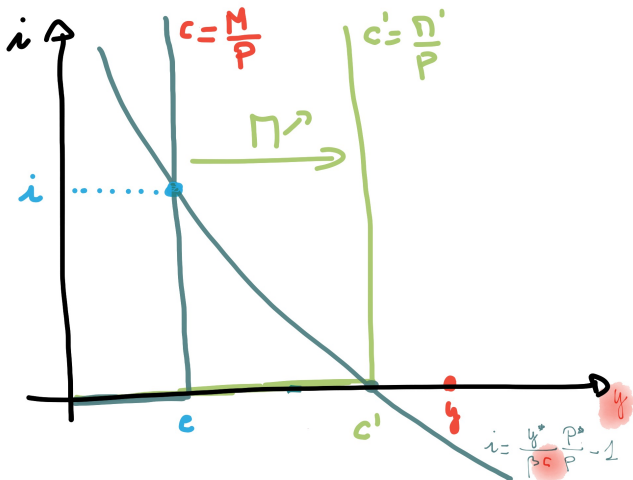
Equilibrium with fix prices - Period 0 - The liquidity trap

- ▶ Now assume that $M = M'$ such that $i = 0$
- ▶ The economy enters a liquidity trap
- ▶ The CIA is not binding anymore : the money increases will not decrease i
- ▶ The economy is stuck at $c < y$ and monetary policy is inefficient.
- ▶ For a money supply M'' , no increase in c

3. A Microfounded Model

Equilibrium with fix prices - Period 0 - The liquidity trap

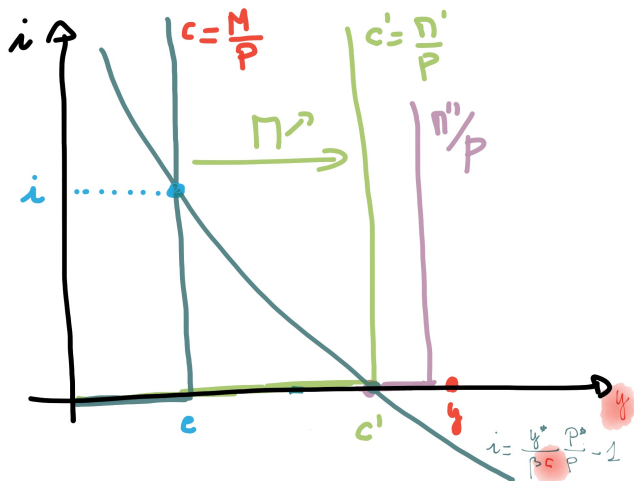
FIGURE 36 : Equilibrium in period 0 - Fix prices



3. A Microfounded Model

Equilibrium with fix prices - Period 0 - The liquidity trap - Monetary policy is inefficient

FIGURE 37 : Equilibrium in period 0 - Fix prices



4. Efficient policy in the liquidity trap

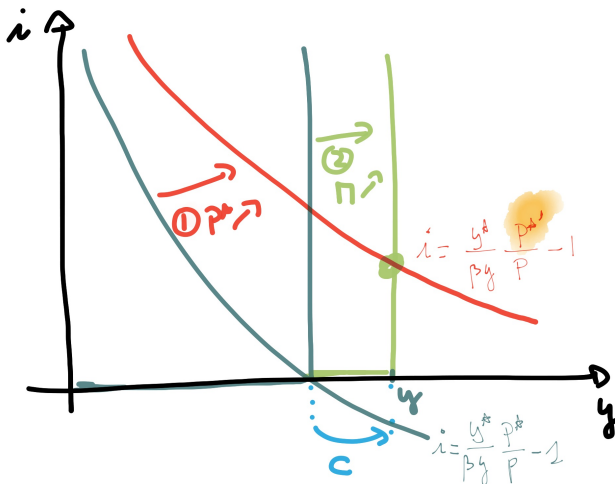
Manipulating expectations : “committing to be irresponsible”

- ▶ If the Central Bank can raise the inflation expectations of the agents (P^*/P) by being credible to increase M^* tomorrow,
- ▶ Then it increases the nominal interest rate today,
- ▶ and it is therefore possible to increase M and close the gap between c and y .

4. Efficient policy in the liquidity trap

Manipulating expectations : "committing to be irresponsible"

FIGURE 38 : Equilibrium in period 0 - Fix prices - increasing M and committing to increase M^*



4. Efficient policy in the liquidity trap

Fiscal policy

- ▶ Assume now that the government taxes agents using lump sum tax and spends on goods, g in period 0 and g^* after.
- ▶ Consider first the flex price case
- ▶ Total demand is now $y = c + g$, and consumption is therefore $c = y - g$
- ▶ The equilibrium equations become :

$$i = \frac{y^* - g^*}{\beta(y - g)} \frac{P^*}{P} - 1$$

$$P = \frac{M}{y} \quad \text{if } i > 0$$

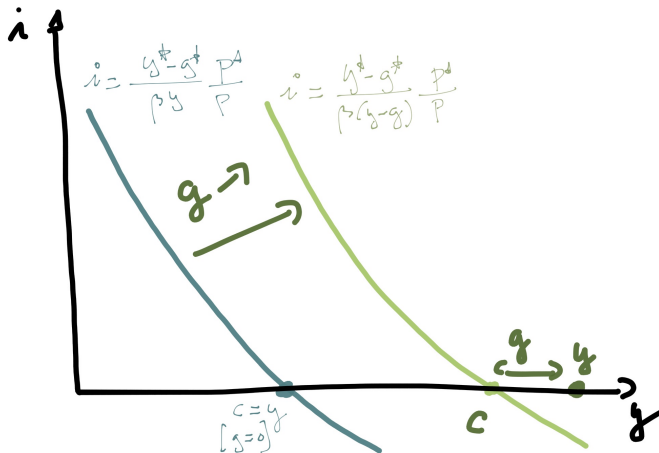
$$P = \bar{P} \quad \text{if } i = 0$$

- ▶ When $g > 0$, the nominal interest curve is shifted up.

4. Efficient policy in the liquidity trap

Fiscal policy

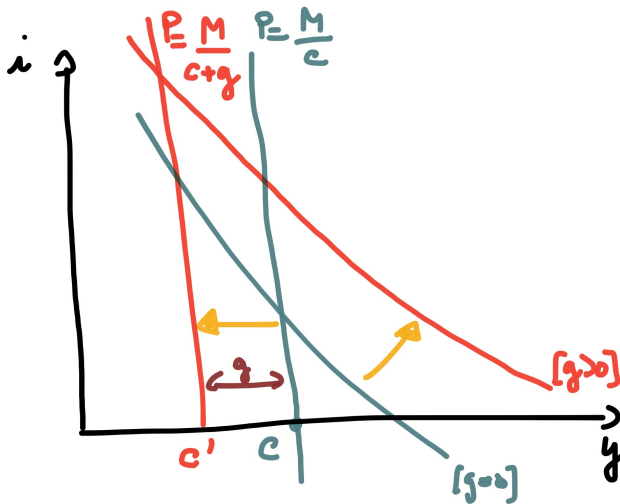
FIGURE 39 : Equilibrium in period 0 - Euler Equation - comparison between $g = 0$ and $g > 0$



4. Efficient policy in the liquidity trap

Fiscal policy

FIGURE 40 : Equilibrium in period 0 - Flex Price - comparison between $g = 0$ and $g > 0$



4. Efficient policy in the liquidity trap

Fiscal policy

- ▶ Assume that the economy is in the liquidity trap with fix prices
- ▶ Fiscal policy will push up the nominal interest rate by decreasing c
- ▶ This will make the CIA constraint binding and c will increase.

4. Efficient policy in the liquidity trap

Fiscal policy

FIGURE 41 : Equilibrium in period 0 - Fix prices - Fiscal policy in the liquidity trap

